Lab 2.2: Implementing Exception Handling

In this lab, you will handle exceptions thrown by **CFileException** in an application.

To see a demonstration of the solution for this lab, click this icon.
(CD-ROM plays the demonstration, "Lab 2.2 Demonstration.")

Estimated time to complete this lab: **15 minutes**

To complete the exercises in this lab, you must have the required software. For detailed information about the labs and setup for the labs, see Labs in this course.

The code that forms the starting point for this lab is located in the folder \Labs\Ch02\Lab2.2\Ex01.

The solution code for this lab is located in the folder \Labs\Ch02\Lab2.2\Ex01\Solution.

Objectives

After completing this lab, you will be able to:

- Use **try** and **catch** blocks to handle exceptions in MFC code.

Prerequisites

There are no prerequisites for this lab.

Exercises

The following exercise provides practice working with the concepts and techniques covered in this chapter.

Exercise 1: Handling Exceptions

In this exercise, you will handle exceptions thrown by **CFileException**. You will add exception-handling code to an application that enables you to choose files to compare in the two panes of a splitter window.

40 Exercise 1: Handling Exceptions

The code that forms the starting point for this exercise is in \Labs\Ch02\Lab2.2\Ex01.

In this exercise, you will handle exceptions thrown by **CFileException**. You will add exception-handling code to an application that enables you to choose files to compare in the two panes of a splitter window.

▶ **Add exception-handling code**

☐ 1. In the **RunComparison** function of DiffDoc.cpp, set up a **try** block for the program statements that open the file lpszFile1 for reading and serialize a **CDiffView** object as shown:

```
try
{
CFile file(lpszFile1, CFile::modeRead);
CArchive ar(&file, CArchive::load);
pView->Serialize(ar);
}
```

Note Only the overloaded constructor of the **CFile** class, which accepts two parameters, throws an exception if an error occurs. If the file cannot be opened, an exception is thrown, which is caught by the **CFileException** handler.

☐ 2. Place exception-handling code in a **catch** block as follows to determine why the open operation failed in the event that a failure occurs:

```
catch(CFileException* e)
{
    if(e->m_cause == CFileException::fileNotFound)
    {
        char errmsg[80];
        strcpy(errmsg,"File  ");
        strcat(errmsg, lpszFile1);
        strcat(errmsg,"  not found");
        ::MessageBox(0,errmsg,"Error",MB_OK);
    }
    else if(e->m_cause == CFileException::tooManyOpenFiles)
        ::MessageBox(0,"No more file handles available","Error",MB_OK);
    else
        ::MessageBox(0,"Open failed","Error",MB_OK);
    e->Delete();
}
```

The code in the **catch** block runs only if the code within the **try** block throws an exception of the type specified in the catch statement.

☐ 3. Repeat steps 3 and 4 for the file lpszFile2.

☐ 4. Save DiffDoc.cpp.

Notes

To see an example of how your code should look, click this icon.

(CD-ROM displays the following sample code.)

```
void CDiffDoc::RunComparison (LPCSTR lpszFile1, LPCSTR lpszFile2)
{
    CMainFrame * pFrame= CDiffApp::GetApp()->GetMainFrame();
    if(pFrame)
    {
        CSplitter * pSplitter= pFrame->GetSplitter();
        if(pSplitter)
        {
            CDiffView * pView;
            pView = (CDiffView *)pSplitter->GetPane(0,0);

            if (pView)
            {
                try
                {
                CFile   file(lpszFile1, CFile::modeRead);
                CArchive ar(&file, CArchive::load);
                pView->Serialize(ar);
                }
                catch(CFileException* e)
                {
                if(e->m_cause == CFileException::fileNotFound)
                {
                    char errmsg[80];
                    strcpy(errmsg,"File  ");
                    strcat(errmsg, lpszFile1);
                    strcat(errmsg,"  not found");
                    ::MessageBox(0,errmsg,"Error",MB_OK);
                }
                else if(e->m_cause == CFileException::tooManyOpenFiles)
                    ::MessageBox(0,"No more file handles
available","Error",MB_OK);
                else
                    ::MessageBox(0,"Open failed","Error",MB_OK);
                e->Delete();
                }
            }
```

(Code continues on following page.)

Notes

```
            pView = (CDiffView *)pSplitter->GetPane(0,1);
            if (pView)
            {
                try
                {
                CFile   file(lpszFile2, CFile::modeRead);
                CArchive ar(&file, CArchive::load);
                pView->Serialize(ar);
                }
                catch(CFileException* e)
                {
                if(e->m_cause == CFileException::fileNotFound)
                {
                    char errmsg[80];
                    strcpy(errmsg,"File  ");
                    strcat(errmsg, lpszFile2);
                    strcat(errmsg,"  not  found");
                    ::MessageBox(0,errmsg,"Error",MB_OK);
                }
                else if(e->m_cause == CFileException::tooManyOpenFiles)
                    ::MessageBox(0,"No more file handles
available","Error",MB_OK);
                else
                    ::MessageBox(0,"Open failed","Error",MB_OK);
                e->Delete();
                }
            }
            //      Flag as clean so that we won't get
            //      prompted to save
            SetModifiedFlag(FALSE);
        }
    }
}
```

For more information about exception handling, see Exception-Handling Process in this chapter.

▶ **Build and test the application**

- Run your application and try to open a file that does not exist.

The solution code for this exercise is located in the folder \Labs\Ch02\Lab2.2\Ex01\Solution.

Notes